MICROPULSE COACH GUIDE

GPS Load on Rest Days

How the system handles match and training carryover

Edition: May 2026 · v1

MicroPulse · Performance Intelligence

1. Why this guide exists

You have probably noticed that the morning after a match, your squad shows up on the dashboard with high values: PL 2.18×, mechanical fatigue, high composite. That makes sense — they just played. But why does this not fade after 2-3 rest days?

Until recently the system had a small bug that made match carryover look like today's load — even when the match was 3 days old. This guide explains how the system now handles this correctly and what it means for your daily decision-making.

You don't need any technical background to read this. Everything is explained with real-world examples from match weeks you recognize.

2. What these numbers actually mean

PL spike (Player Load multiplier)

Player Load is a direct measurement from the accelerometer in the Catapult pod — it captures how hard the body worked physically during the session or match (running, jumping, changes of direction, contact). **PL spike** is how many times higher (or lower) today's Player Load is compared to the player's 28-day baseline.

Example:

- Andri — 28-day baseline: 275 AU
- Sunday match: 600 AU
- PL spike = 600 / 275 = 2.18×

So 2.18× means: Andri worked 2.18 times harder than usual on that specific day.

Composite Load (Comp 0.78)

Composite is a **combined** score that blends several signals into one 0-1 metric:

- 55% — internal load (RPE × duration ACWR over the last 7 days)
- 45% — external load (GPS spikes: PL, HIR, decel, accel)
- + Residual MLI sanity check
- + Decel Burden (McBurnie module)

Color guide:

Color	Comp value	What it means
Green	0.00 - 0.15	No concern — player is recovered
Light green	0.15 - 0.40	Mild concern — monitor in next session
Yellow	0.40 - 0.68	Moderate load — consider modifications
Red	0.68 - 1.00	High load — cap or modify the plan

Mechanical fatigue tag

This tag appears when GPS spikes (PL, HIR, decel) are consistently above threshold. It is the system's way of saying: this player is under mechanical overload. Before the fix went in, this tag could appear on MD+3 because of a match on MD+0 — which was wrong.

3. The problem we had

Before the fix went in (May 2026), it looked like this:

Scenario: Sunday match, MD+1 and MD+2 rest, MD+3 training

Day	What the system saw	What the system displayed
Sun (MD)	Andri plays the match. PL 600 AU.	PL 2.18× - high comp, mechanical fatigue. CORRECT.
Mon (MD+1)	No training. No GPS upload.	PL 2.18× - system fell back to Sun row as 'today'.
Tue (MD+2)	No training. No GPS upload.	PL 2.18× - same Sun row again.
Wed (MD+3) AM	Training today, not uploaded yet.	PL 2.18× mechanical fatigue - WRONG.

The system kept showing the Sunday match as 'current load' all week. The coach reading the dashboard on Wednesday morning thought: Andri is still under stress, maybe a lighter session today — but in reality Andri had recovered.

Why the system did this

The intent was correct in theory: if there is no GPS row for today (because training has not happened yet), the dashboard should not be blank. The system fell back to the most recent known GPS row so the coach had something to look at.

The problem was that it didn't distinguish between 'fresh data from today' and 'old data from Sunday's match'. The coach read both as the same thing.

4. How the system works now

The system now applies recovery decay to carryover data — based on research by Hader and colleagues 2019 in professional football.

Hader 2019 recovery curve

Hader measured muscle damage (CK), neuromuscular performance (CMJ), heart rate variability (HRV) and perceived fatigue across 5 days following a 90-minute official soccer match in 32 elite players. The result was:

Days post-match	% of load still present	Sport-science interpretation
D+0 (same day)	100%	Full neuromuscular fatigue, muscle damage peaks
D+1	55%	Acute fatigue dominant — light sessions
D+2	30%	Mostly recovered, residual stiffness
D+3	15%	Essentially baseline — normal training fine
D+4	5%	Negligible
D+5+	0%	Pure baseline

What this looks like for the Andri example

Day	Raw Sun spike	Decay	Shown on dashboard
Sun (MD)	2.18×	1.00	2.18× — CORRECT, match is today
Mon (MD+1)	2.18×	0.55	1.20× — some residual fatigue, yellow
Tue (MD+2)	2.18×	0.30	0.65× — mostly back, below flag
Wed (MD+3)	2.18×	0.15	0.33× — GREEN, ready to train
Wed (MD+3) PM	—	—	After evening session uploads: real Wed load, not carryover

> So on Wednesday morning you read Andri as GREEN and ready. You can plan a normal session without falsely-flagged carryover holding you back.

5. What this means for your daily routine

MD+0 (matchday)

Spike values after the match appear unchanged. You read the actual match load. No decay on the actual match day.

Practical: check compliance (did anyone forget to turn the pod on?), HSR exposure distribution, and who played the most minutes.

MD+1 and MD+2 (recovery days)

Spike values get decay applied (0.55× and 0.30× respectively). So a player who had 2.18× on Sun shows 1.20× on Mon and 0.65× on Tue.

Practical: yellow zone on MD+1 is normal — this is 'haven't fully recovered yet'. By MD+2 most should be in lower yellow or green. If a player is still red on MD+2, check the wellness check-in (sleep, soreness, stress) to understand why.

MD+3 and beyond (back to training)

Spike values are near zero (decay 0.15 and lower). The squad should be GREEN unless something else is going on.

Practical: if someone is still red on MD+3 it's likely:

- High Foster Strain (last 7 days RPE-based — not affected by GPS decay)
- Personal baseline issue (they have been undertrained for a long time)
- Recorded injury in the system
- Poor sleep / high soreness in today's check-in

> Important: GPS decay is about GPS values. RPE, wellness, and injury status are independent pipelines. If Foster Monotony reports 7 days of high load, that is still a legit signal — not affected by decay.

Live training day

When training takes place and GPS is uploaded, the system switches from decayed-fallback to today's real-time data. No more decay — you read the actual load of the day.

6. When to still pay attention

The decay model is built on averages — it is correct in 80-90% of cases. The following situations can be exceptions:

Match congestion (2 matches per week)

The decay model assumes the next session is a training. If there is another match 3 days after the previous one, fresh match load lands on top of the older carryover. The system has a separate Residual MLI / Decel module that captures this — you don't need to calculate yourself.

Older players (35+)

Hader 2019 was primarily on players aged 23-32. Older players (or youth under 18) have slower recovery. The decay then becomes optimistic. Practical: read their wellness check-in (sleep, soreness) particularly carefully on MD+2.

Unusually long matches

90 minutes is the standard. If there was extra time + penalties (120 minutes), the decay is slightly too fast. Plan to need 1 extra day.

Poor sleep / illness

The wellness check-in catches this. If a player reports 4 hours of sleep and high soreness on MD+2, readiness score is low even if GPS spike has gone green. Trust wellness in this case.

Recorded injuries

Injury status (injured, rehab, RTP) is set manually by you. It always overrides the verdict. If a player got injured in the Sunday match and is on rehab status, they don't qualify as 'ready to train' regardless of what GPS decay says.

7. What this does NOT touch

This fix only affects GPS spike ratios. The following pipelines work exactly as before:

Signal	How it works	Affected by GPS decay?
RPE-based ACWR (Foster)	Read from session_rpe_entries 7-day rolling. Each day has its own entry.	No — RPE is never 'fallback'
Foster Monotony	Mean / SD of 7-day RPE	No
Foster Strain	Weekly RPE × Monotony	No
Wellness check-in	Daily input from player (sleep, fatigue, stress, soreness, energy)	No — manual every day
VALD jump tests	Timestamp-explicit in the data	No
Injury status	Coach-managed state	No — always overrides
Internal:External decoupling	Requires both GPS + RPE on same day	Only active on real days

8. 30-second summary

Before: the system reused Sunday match data and labelled it as today's stress all week. Coach read a false alarm on MD+3. Now: the system applies Hader 2019 recovery decay (1.0 / 0.55 / 0.30 / 0.15 / 0.05) to carryover GPS values. PL 2.18× on Sun becomes 0.33× on Wed — which is correct given 2 rest days. Impact on your work: the dashboard is now trustworthy on rest mornings. You get GREEN status on MD+3 for players who have actually recovered. The mechanical fatigue tag only appears when it is real.

9. Scientific references

This decay model is based on the following research:

Hader, K., Rumpf, M. C., Hertzog, M., Kilduff, L. P., Girard, O., & Silva, J. R. (2019). Monitoring the Athlete Match Response: Can External Load Variables Predict Post-Match Acute and Residual Fatigue in Soccer? A Systematic Review. Sports Medicine - Open, 5(1), 48.

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Silva, J. R., Rumpf, M. C., Hertzog, M., Castagna, C., Farooq, A., Girard, O., & Hader, K. (2018). Acute and Residual Soccer Match-Related Fatigue: A Systematic Review and Meta-analysis. Sports Medicine, 48(3), 539-583.

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